

Week 03 Task 9

Network Traffic Analysis — Wireshark Basics

CandidateName: Asad Ali
Reference Number: TBS-INT-2026-445
Submission Date: 3|11|2026
Submitted To: TechBiz Security

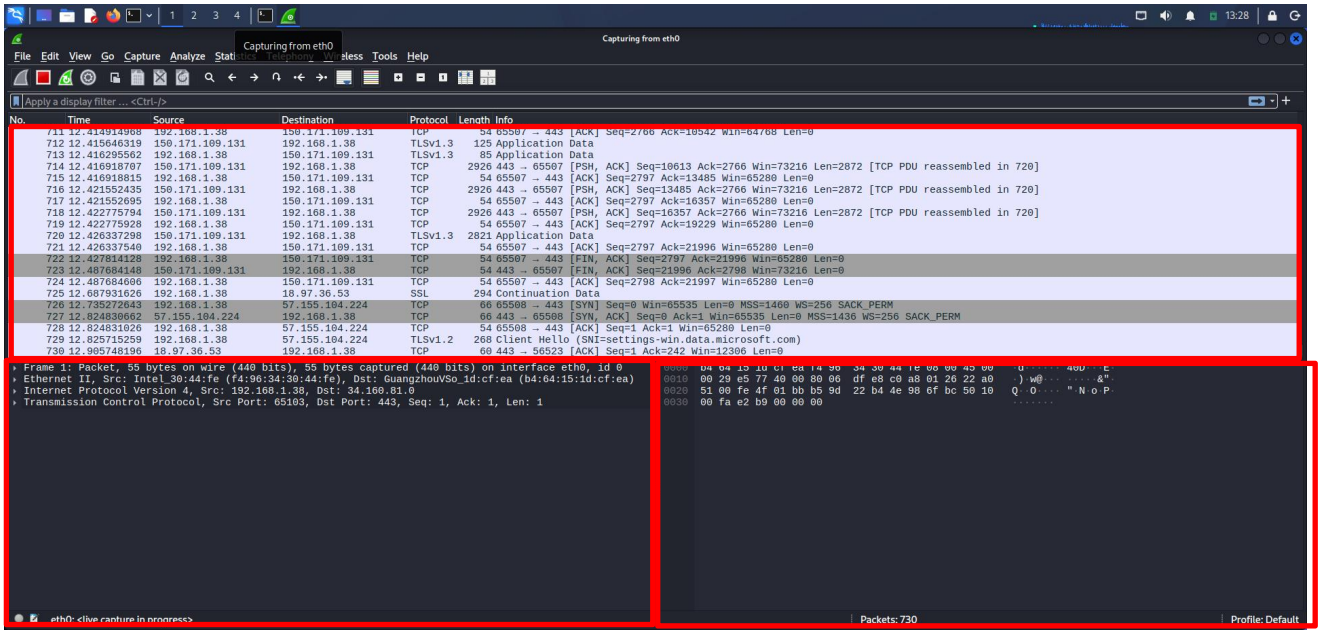
Table of Contents

1) Executive Summary 3
2) Interface of Wireshark 3
3) Filters of Wireshark 3
4) Inspect One HTTP Request Packet 5
5) Understanding how something suspicious look in Wireshark 6
6) TryHackMe Room Traffic Analysis Essentials 7
7) Conclusion 7

1) Executive Summary

This task focused on developing practical network traffic analysis skills using Wireshark. Initially, I explored the Wireshark interface and its main components, including the packet list, packet details, and packet bytes panes. I then demonstrated the use of display filters such as **dns, http**, and **ip.addr** to isolate and analyze specific network traffic. Afterward, I inspected an individual HTTP GET request packet, examining its request method, URI, Host header, User-Agent, and other HTTP headers. I also studied how potentially suspicious network activities can appear in Wireshark, including unusual communication patterns and unexpected traffic behavior. Finally, I completed the TryHackMe **Traffic Analysis Essentials** room to strengthen my understanding of packet inspection, network protocols, and identifying potentially malicious network activity.

2) Interface of Wireshark

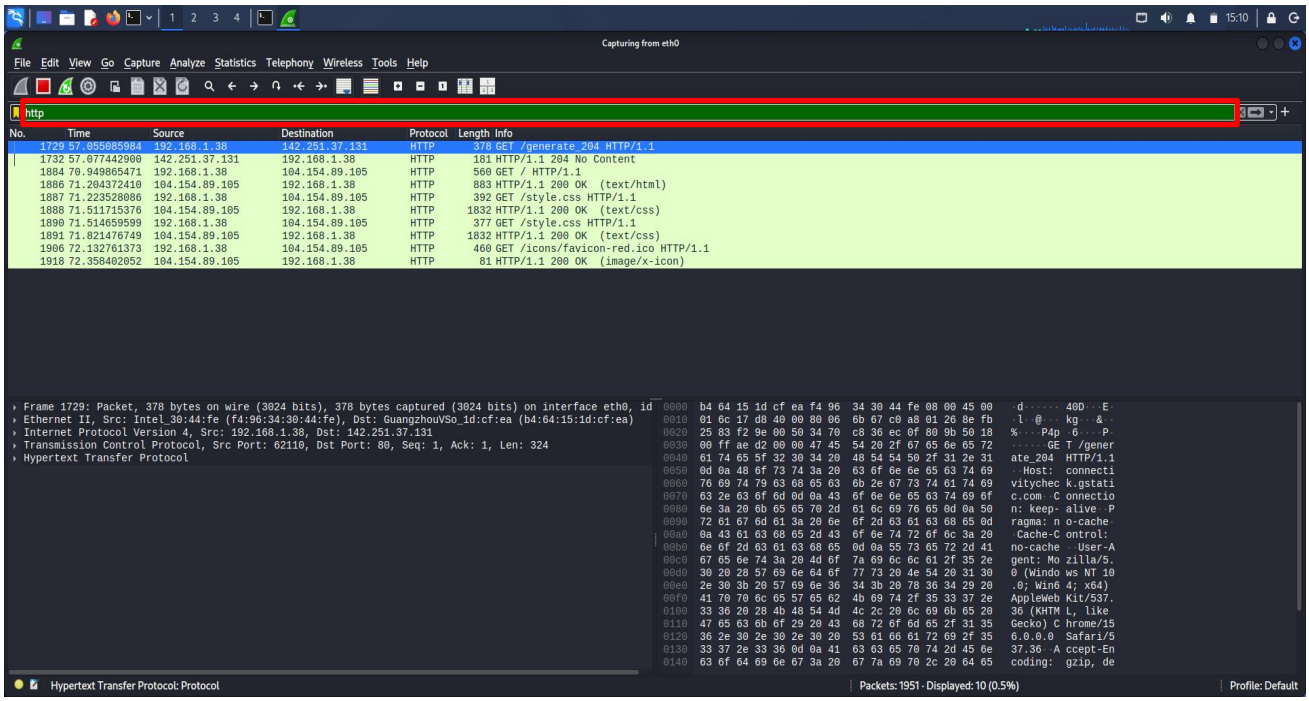


The above figure shows the Wireshark main interface during a live capture on eth0, with the menu bar, toolbar and an empty display filter bar at the top. The packet list pane shows each captured packet's number, time, source, destination, protocol, length and info. The packet details pane breaks the selected frame into layers (Ethernet II, IPv4, TCP), while the packet bytes pane shows the raw hex and ASCII data. The status bar shows 730 packets captured.

3) Filters of Wireshark

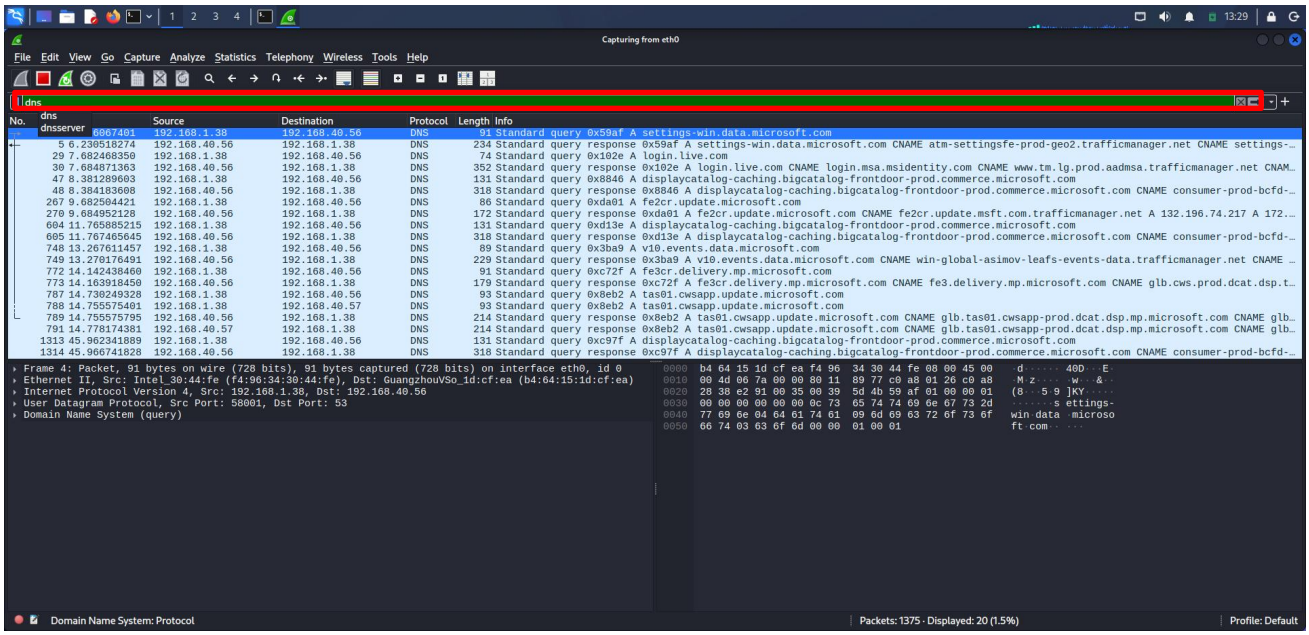
Wireshark filters are rules or expressions used to sort, isolate, and remove noise from network packet traces so you can view only the specific traffic that matters to your investigation. There are two types of wireshark filters. **Capture Filters** and **Display Filters**. Capture filters applied before or during live capturing of traffic whereas Display filters are packets have already been captured or loaded into wireshark.

3.1) Http



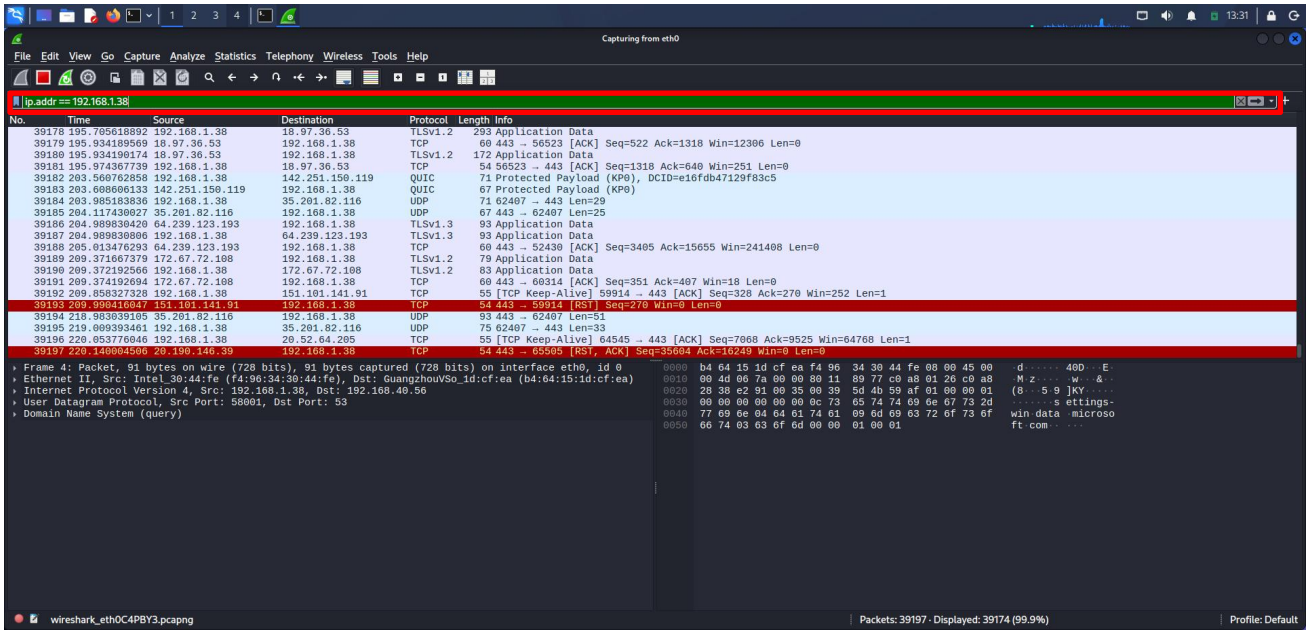
In the above figure it is shown that The http display filter is use to isolate HTTP packets from the captured traffic, making it easier to inspect web communication.

3.2) dns



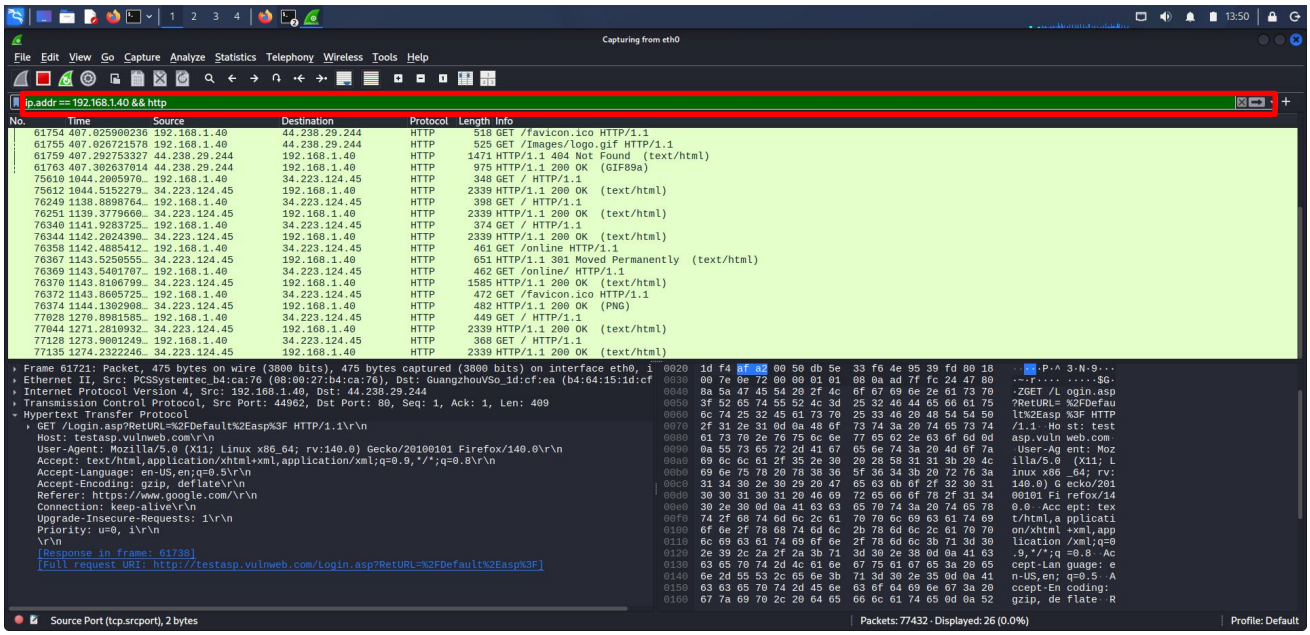
In the above figure it is shown that dns filter is used. DNS traffic isolated showing standard queries and responses for Microsoft domains.

3.3) ip.addrr



The above figure The demonstrates filtering network traffic using an IP address to isolate packets associated with a specific host for further analysis

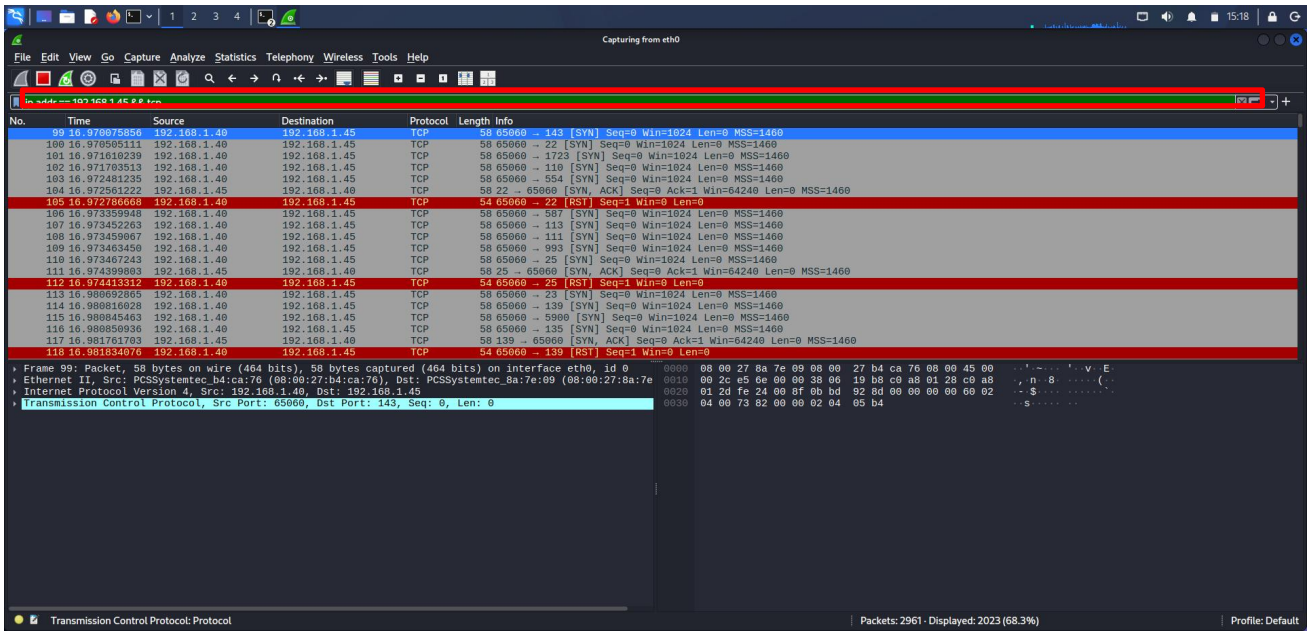
4) Inspect One HTTP Request Packet



In the above figure it is shown that I inspected the http header and get the information explained below.

- **Request line:** GET /Login.asp?RetURL=%2FDefault%2Easp%3F HTTP/1.1
- **Host:** testasp.vulnweb.com
- **User-Agent:** Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0
- **Other headers:** Accept, Accept-Language, Accept-Encoding, Referer: https://www.google.com/, Connection: keep-alive, Upgrade-Insecure-Requests: 1 and Priority: u=0, i.
- **Packet details:** the request goes from 192.168.1.40:44962 to 44.238.29.244:80 over TCP, with a response in frame 61738 and a full request URI shown.
- **Filter:** ip.addr == 192.168.1.40 && http is valid.

5) Understanding how something suspicious look in Wireshark

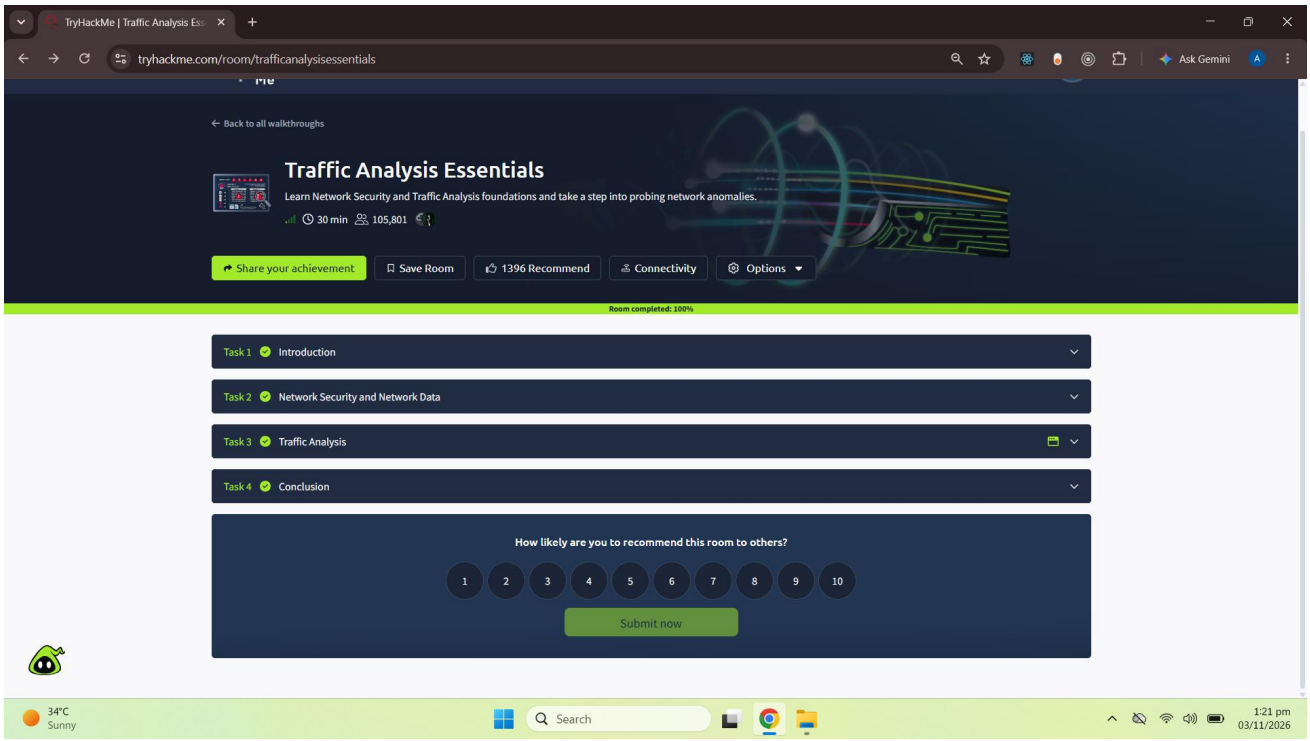


The above screenshot demonstrates potentially suspicious TCP traffic between the source host `192.168.1.40` and the destination host `192.168.1.45`. The packet list shows the source attempting to connect to multiple destination ports, including ports 22, 25, 110, 139, and 5900, within a very short period. These connection attempts are represented by TCP SYN packets, indicating attempts to establish connections with different services on the target host.

The traffic appears suspicious because rapidly probing multiple ports on a single host is consistent with TCP port scanning, a reconnaissance technique used to identify accessible services. Several attempts receive TCP RST responses, indicating that the target is resetting the connection attempts, often because the ports are closed. This systematic pattern differs from typical application communication and may indicate automated scanning activity. However, further investigation is required to determine whether the activity was authorized or malicious.

6) TryHackMe Room Traffic Analysis Essentials

TryHackMe Room Wireshark the Basics is premium that's why I utilized the room **Traffic Analysis Essentials** as shown in Image below.



7) Conclusion

This task provided practical experience in network traffic analysis using Wireshark. I learned to navigate its interface, apply capture and display filters, and isolate specific traffic using HTTP, DNS, and IP address filters. I also inspected an HTTP GET request to understand its request method, headers, source, and destination information.

Furthermore, I developed an understanding of how potentially suspicious network activity can be identified by examining unusual communication patterns and unexpected traffic behavior. Completing the TryHackMe Traffic Analysis Essentials room reinforced my knowledge of packet inspection and network protocols. Overall, this task strengthened my foundational network analysis skills and improved my ability to examine traffic and recognize activity that may require further investigation.